

Trend Analysis of India's Arms Imports (2020–2026)

Chapter 93 (ITC-HS): An Analysis Under the Atmanirbhar Bharat Initiative

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1 Introduction

This note analyses monthly import data for ITC-HS Chapter 93 (Arms and Ammunition) over the period April 2020 to March 2026, covering 14 six-digit ITS(HS) codes. The dataset is sourced from DGCIS portal and covers the period marked by India’s Positive Indigenisation List (PIL) notifications under the Atmanirbhar Bharat defence self-reliance initiative.

Total arms imports over the six-year period amounted to approximately **\$671.3 million**, with a monthly average of \$9.3 million. The distribution is highly skewed: HS 930690 (Bombs, Grenades, and Other Munitions alone accounted for **64.3%** of total import value, followed by HS 930599 (Parts of Heading 9302 and 9304) at 13.2% and HS 930400 (Other Arms) at 7.3%.

2 Data Description

2.1 HS Code Coverage

The dataset covers the following 14 six-digit HS codes under Chapter 93:

Table 1: Chapter 93 HS Codes: Import Share (Apr 2020 – Mar 2026)

HS Code	Description	Value (USD)	Share (%)
930690	Bombs, Grenades and Other Munitions	431,944,753	64.3
930599	Parts of Heading 9302 and 9304	88,857,253	13.2
930400	Other Arms (Air/Gas Guns etc.)	48,841,991	7.3
930621	Shotgun Cartridges	38,177,950	5.7
930630	Other Cartridges and Parts	23,666,140	3.5
930629	Shotgun Parts and Air Gun Pellets	13,177,183	2.0
930520	Parts/Accessories of Shotguns/Rifles	8,622,393	1.3
930330	Sporting/Hunting/Target-Shooting Rifles	5,003,921	0.7
930320	Sporting/Hunting/Target-Shooting Shotguns	4,587,752	0.7
930200	Revolvers and Pistols	4,307,749	0.6
930390	Other Firearms and Similar Devices	2,074,773	0.3
930510	Parts/Accessories of Revolvers/Pistols	1,902,967	0.3
930310	Muzzle-Loading Firearms	109,543	0.0
930700	Swords, Bayonets and Similar Arms	28,784	0.0
Total		671,303,152	100.0

3 Time Series Analysis

3.1 Raw Import Series

Figure 1 plots the monthly aggregate import value (summed across all 14 HS codes) from April 2020 to March 2026. Several features are immediately apparent:

- **Near-zero imports (Apr–Sep 2020):** The first six months show negligible import activity, attributable to COVID-19 induced disruptions in global logistics, port closures, and deferral of procurement schedules.
- **Sharp escalation from Oct 2020:** Imports surge markedly from October 2020 onward, coinciding with the post-Galwan Valley clash emergency procurement authorisations and the reopening of global supply chains.
- **Persistent lumpiness:** The series does not follow a smooth trend. Extremely large one- or two-month spikes separated by low baseline months are the defining characteristic, consistent with the contract-milestone-driven nature of Indian defence procurement under the Defence Acquisition Procedure (DAP).

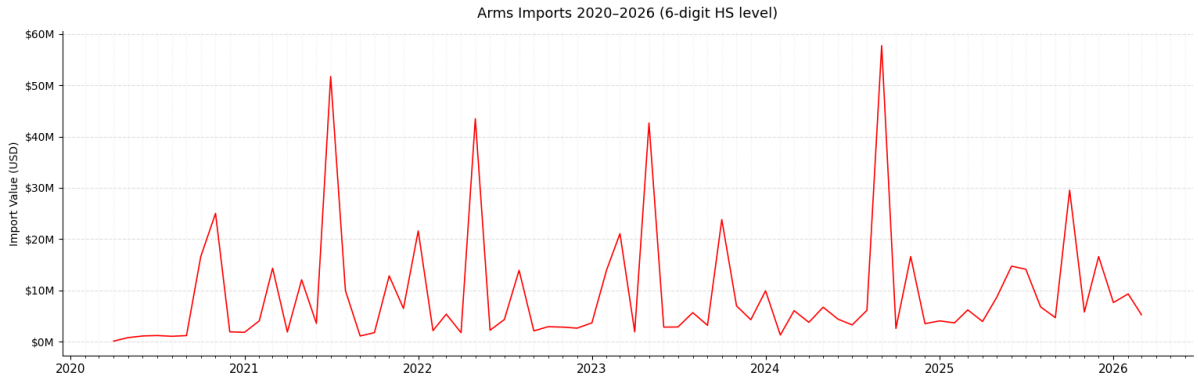


Figure 1: Monthly Arms Imports, India (Apr 2020–Mar 2026), Aggregate of 14 Six-Digit HS Codes under Chapter 93. Y-axis in USD millions.

3.2 Time Series Decomposition

Using the *statsmodels* library in python, we have separated the underlying long-run trend from seasonal patterns and random shocks, an additive decomposition is applied with a period of 12 months:

$$M_t = T_t + S_t + \varepsilon_t \quad (1)$$

where M_t is the observed import value in month t , T_t is the trend component (extracted via centred moving average), S_t is the seasonal component, and ε_t is the irregular residual.

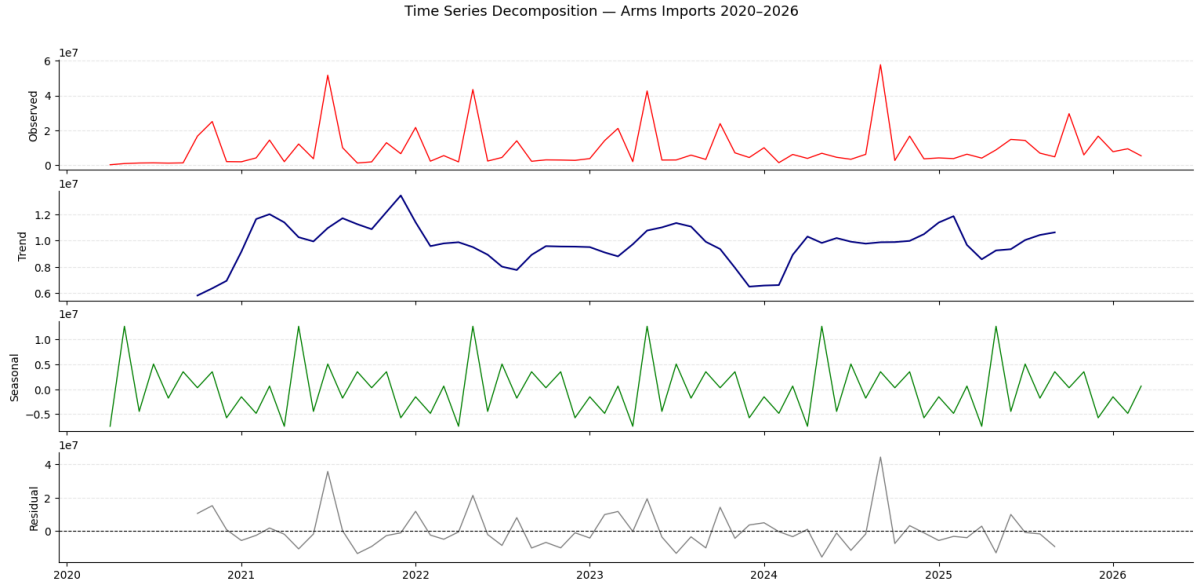


Figure 2: Additive Time Series Decomposition of Monthly Arms Imports (Apr 2020–Mar 2026). Panels show (top to bottom): observed series, trend, seasonal component, and residual. All axes are in USD.

3.2.1 Trend Component

The trend panel (Figure 2, second panel) reveals three distinct phases:

1. **Rising phase (early 2021 – mid 2022):** The trend rises from approximately \$6M per month in early 2021 to a peak of \$13M by mid-2022. This reflects the post-COVID procurement catch-up and emergency acquisitions in the wake of border tensions.
2. **Declining phase (mid 2022 – mid 2024):** The trend declines from \$13M to approximately \$6M by mid-2024. This coincides with the progressive notification of Positive Indigenisation Lists (PIL-1: August 2020; PIL-2: May 2021; PIL-3 and beyond: 2022–2023), which banned imports of listed items, exerting downward pressure on the import trend.
3. **Recovery phase (mid 2024 – 2026):** The trend recovers to approximately \$10M by early 2026, potentially reflecting procurement of non-indigenised items, parts, and ammunition categories not covered by PIL restrictions.

3.2.2 Seasonal Component

The seasonal panel shows a **strong and recurring** intra-year pattern. Positive seasonal effects concentrate in **July** and **October–November** each year, with negative effects in **April** and **August–September**. This pattern is consistent with:

- Government of India fiscal year procurement cycles (budget releases in Q1 and Q2 driving mid-year purchasing)
- Year-end budget utilisation pressure generating February–March clearance spikes in some years

3.2.3 Residual Component

The residual panel isolates genuine shocks not explained by trend or seasonality. Two large positive residuals are visible:

- **July 2021** ($\varepsilon_t \approx +\$38\text{M}$ above trend+seasonal)
- **September 2024** ($\varepsilon_t \approx +\$43\text{M}$ above trend+seasonal)

These represent true anomalous procurement events requiring policy-level explanation (see Section 4).

4 Spike Analysis

4.1 Identified Spikes

Table 2 lists the ten largest monthly import values identified from the data.

Table 2: Top 10 Monthly Import Spikes, Chapter 93 (Apr 2020–Mar 2026)

Rank	Month	Value (USD)	Probable Cause
1	Sep 2024	\$57,750,762	Large platform delivery milestone; possible front-loading ahead of PIL extension
2	Jul 2021	\$51,716,272	PIL-2 anticipation front-loading (PIL-2 notified May 2021); post-Galwan emergency stocks
3	May 2022	\$43,471,941	Russia–Ukraine disruption; accelerated deliveries from alternative suppliers
4	May 2023	\$42,648,308	Contract milestone payments; ammunition (9306.90) bulk procurement
5	Oct 2025	\$29,537,410	FY budget utilisation; new procurement authorisations
6	Nov 2020	\$25,014,557	Post-Galwan emergency procurement; first PIL anticipation effect
7	Oct 2023	\$23,809,781	Contract bunching; FY Q2 procurement cycle
8	Jan 2022	\$21,596,793	Delayed FY 2021–22 deliveries; post-holiday customs clearance
9	Mar 2023	\$21,067,073	FY-end budget utilisation pressure
10	Nov 2024	\$16,595,983	Follow-on deliveries from Sep 2024 contract

4.2 Attribution of Major Spikes

4.2.1 Nov–Dec 2020: Post-Galwan Emergency Procurement

The Galwan Valley clash (June 2020) triggered emergency procurement authorisations outside normal DAP timelines. Fast-track acquisitions of small arms, crew-served weapons, and munitions (primarily HS 9306.90) resulted in a sharp \$25M spike in November 2020. The concurrent notification of PIL-1 (August 2020) may have additionally motivated importers to front-load items before the embargo date.

4.2.2 Jul 2021: PIL-2 Anticipation and Emergency Stocks

The second largest spike (\$51.7M) occurs in July 2021, two months after the notification of PIL-2 (May 2021). This timing is consistent with an *anticipatory front-loading effect*: procurement agents accelerated import orders for items scheduled for indigenisation before PIL-2’s effective restrictions took hold. The dominance of HS 9306.90 in total imports makes munitions the most plausible category driving this spike.

4.2.3 May 2022: Russia–Ukraine Supply Chain Disruption

The onset of the Russia–Ukraine conflict (February 2022) created acute uncertainty in Russia-origin defence supply chains. India, which sources a significant share of its military hardware from Russia, likely front-loaded pending deliveries before logistics constraints tightened under Western sanctions. The \$43.5M spike in May 2022 is consistent with this accelerated delivery hypothesis.

4.2.4 May 2023 and Mar 2023: Contract Bunching

Twin spikes in March and May 2023 (\$21.1M and \$42.6M respectively) within the same financial year suggest milestone-based payment clustering on one or more large ammunition contracts. The March spike likely reflects FY-end budget utilisation; the May spike reflects Q1 delivery commencement under new contracts.

4.2.5 Sep 2024: Largest Single-Month Spike

The largest spike in the dataset (\$57.8M) occurs in September 2024. At more than 6 times the monthly average, this is almost certainly attributable to a single large contract milestone payment, most likely for guided munitions or bomb systems under HS 9306.90. The subsequent follow-on spike in November 2024 (\$16.6M) supports the interpretation of a staged delivery schedule.

5 Year-on-Year Growth Analysis

Table 3 presents average year-on-year growth rates computed as $g_t = (M_t - M_{t-12}) / M_{t-12} \times 100$.

Table 3: Average Year-on-Year Import Growth Rate by Calendar Year

Year	Avg. YoY Growth (%)
2021	+949.4
2022	+96.9
2023	+140.1
2024	+153.9
2025	+166.7

The 2021 figure (+949.4%) is inflated by the near-zero base of April–September 2020 (COVID disruption). Stripping the first two quarters, underlying growth in 2021 was still substantial. Growth remained elevated in 2022–2025, suggesting that PIL restrictions have not yet caused an absolute decline in arms imports at the Chapter 93 aggregate level — only a compositional shift within the chapter (away from PIL-listed items, toward non-listed ammunition and parts).

6 Forecasting Arms Imports

6.1 Stationarity Testing

Before fitting any time series model, it is necessary to establish whether the series is stationary, that is, whether its mean, variance, and autocovariance structure are invariant over time. Non-stationary series produce spurious regression results and violate the assumptions underlying ARIMA-class models.

Two complementary tests are employed, with opposing null hypotheses:

Table 4: Stationarity Tests: Null Hypotheses and Decision Rules

Test	H_0	H_1	Reject H_0 if
ADF	Series is non-stationary	Series is stationary	$p < 0.05$
KPSS	Series is stationary	Series is non-stationary	$p < 0.05$

Running both tests jointly and cross-checking their conclusions avoids the low-power problem of either test in isolation, which is particularly relevant here given the small sample ($T = 72$).

7 Stationarity Test

The stationarity of the time series was assessed using both the Augmented Dickey–Fuller (ADF) test and the Kwiatkowski–Phillips–Schmidt–Shin (KPSS) test.

7.1 Augmented Dickey–Fuller (ADF) Test

- **ADF Statistic:** -9.6607

- **p-value:** 0.0000

Since the p-value is less than the significance level of 0.05, the null hypothesis of the presence of a unit root is rejected. Therefore, the time series is **stationary** according to the ADF test.

7.2 KPSS Test

- **KPSS Statistic:** 0.0794
- **p-value:** 0.1000
- **Lags Used:** 2

Critical Values:

Significance Level	Critical Value
10%	0.3470
5%	0.4630
2.5%	0.5740
1%	0.7390

Since the p-value is greater than 0.05, the null hypothesis of stationarity cannot be rejected. Thus, the KPSS test also indicates that the time series is **stationary**.

7.3 Conclusion

Both the ADF and KPSS tests consistently indicate that the time series is stationary. Therefore, no further differencing is required before fitting time series forecasting models such as ARIMA or SARIMA.

8 Forecasting Methodology and Implementation

8.0.1 Initial Attempt: Auto ARIMA

The first step in model selection was to apply the `auto_arima` procedure from the `pmdarima` library, which performs a stepwise search over the SARIMA order space $(p, d, q)(P, D, Q)_{12}$ and selects The specification minimising the Akaike Information Criterion (AIC). The procedure returned:

$$\text{SARIMA}(0, 0, 0)(0, 0, 0)_{12} \quad (2)$$

This is a **white noise model**, it contains no autoregressive, moving average, or differencing terms in either the seasonal or non-seasonal component. The forecast from such a model provides no economically meaningful information about future import trajectories.

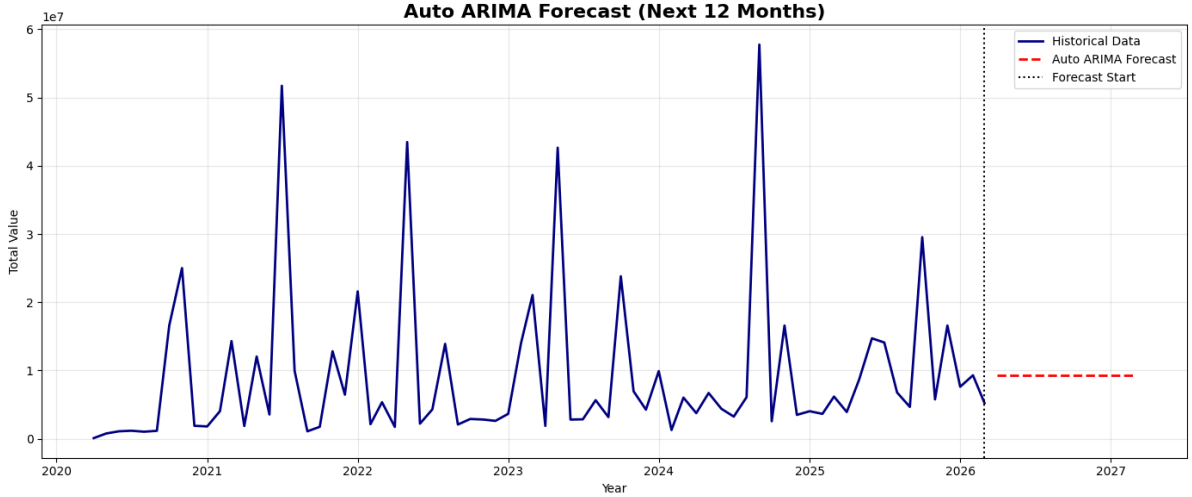


Figure 3: Initial Auto ARIMA Forecast (Next 12 Months) — White Noise Model. The red dashed line shows a flat forecast equal to the sample mean (\$9.3M), produced by the $\text{SARIMA}(0, 0, 0)(0, 0, 0)_{12}$ specification selected by `auto_arima` on the raw spike-contaminated series. This confirms that the algorithm detected no exploitable autocorrelation structure in the raw data.

8.0.2 Diagnosis: Why the White Noise Model Was Selected

The failure of `auto_arima` to identify any ARIMA structure is attributable to the **extreme spike problem** in the data. The series contains several months where import values are 4–6 times larger than the typical monthly value (notably July 2021 at \$51.7M and September 2024 at \$57.8M, against a sample mean of \$9.3M). These outliers have two damaging effects on the model selection procedure:

1. **Autocorrelation destruction:** The spike months dominate the variance of the series. When `auto_arima` computes the sample autocorrelation function (ACF) and partial autocorrelation function (PACF) to determine lag structure, the spike-driven variance swamps the underlying seasonal and trend signal, rendering the ACF and PACF statistically indistinguishable from white noise.
2. **Over-penalisation of complex models:** Since no lag structure is detectable in the spike-dominated ACF, adding AR or MA terms does not improve the likelihood sufficiently to offset the AIC penalty for additional parameters. The algorithm therefore retains the zero-parameter white noise specification.

Figure 4 confirms this diagnosis empirically. Both the ACF and PACF of the raw series show virtually no significant spikes at any lag beyond zero, all autocorrelations fall well within the 95% confidence bounds (shaded blue region). Notably, **lag 12 shows no significant spike in the ACF**, which would normally be expected for a series with strong annual seasonality. This is direct evidence that the extreme procurement spikes have destroyed the autocorrelation structure that genuine seasonal variation would otherwise produce.

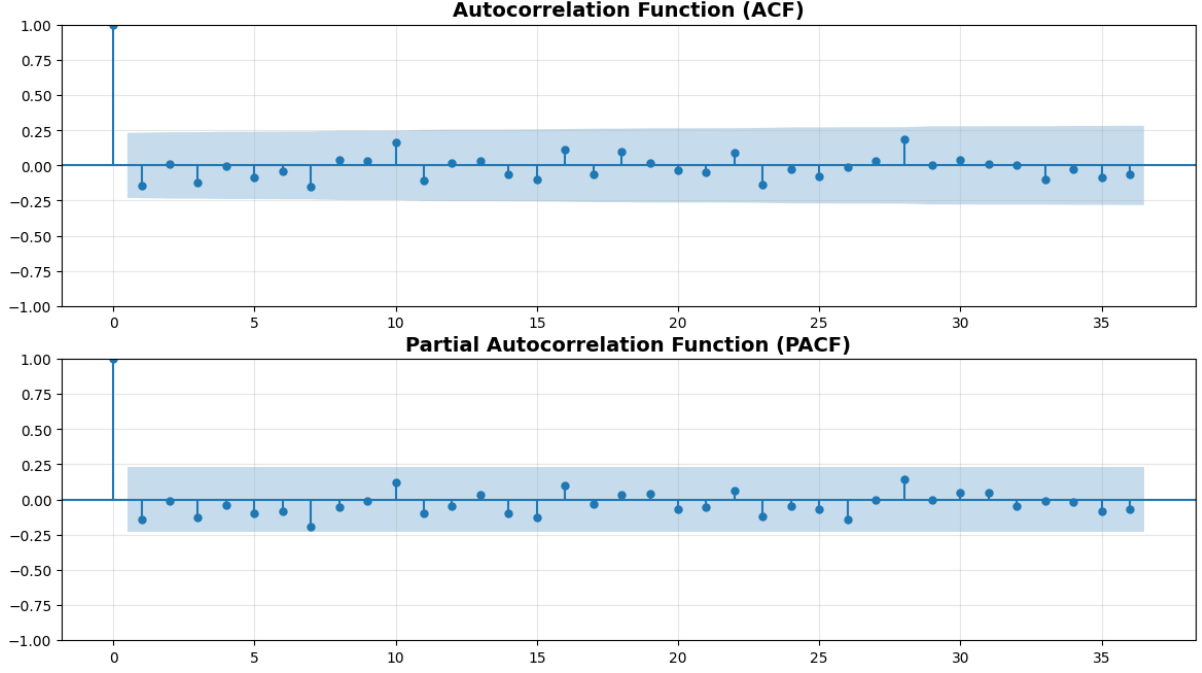


Figure 4: ACF and PACF of Raw Arms Import Series (Apr 2020–Mar 2026). .

This outcome is inconsistent with the visual evidence of strong seasonality in the decomposition (Figure 2), confirming that the problem lies in outlier contamination of the autocorrelation estimates rather than a genuine absence of seasonal structure.

8.0.3 Solution: Winsorisation of Extreme Values

To recover the underlying autocorrelation structure, the series is **winsorised** at the 90th percentile prior to model selection. Winsorisation replaces all values exceeding the 90th percentile threshold $\tau_{0.90}$ with $\tau_{0.90}$ itself:

$$\tilde{M}_t = \min(M_t, \tau_{0.90}), \quad \tau_{0.90} = Q_{0.90}(M_t) \quad (3)$$

where $Q_{0.90}(M_t)$ denotes the 90th percentile of the observed series. For this dataset, $\tau_{0.90} \approx \$15M$, meaning the seven largest spike months are capped at \$15M while all remaining observations are left unchanged.

The rationale for winsorisation rather than deletion is threefold:

1. **Preserves time index:** Deleting spike months would create gaps in the monthly time index, breaking the equally-spaced assumption required by SARIMA estimation. Winsorisation retains all 72 observations at their correct dates.
2. **Recovers seasonal signal:** By attenuating The the spike months to a level closer to the non-spike distribution, winsorisation allows the autocorrelation structure of normal procurement months, which exhibits clear annual seasonality ,to become visible to the model selection algorithm.

3. **Scope is limited to order selection:** Crucially, winsorisation is applied *only* to identify the optimal SARIMA order $(p^*, d^*, q^*)(P^*, D^*, Q^*)_{12}$. The final model is always estimated on the **original unmodified series** M_t , ensuring that forecasts are expressed in true dollar terms and the spike observations remain in the estimation sample.

This two-step procedure can be summarised as:

$$\text{Step 1 (Order selection): } (p^*, d^*, q^*)(P^*, D^*, Q^*)_{12} = \arg \min_{(p,d,q,P,D,Q)} \text{AIC} \left[\text{SARIMA on } \tilde{M}_t \right] \quad (4)$$

$$\text{Step 2 (Estimation): } \text{Fit SARIMA}(p^*, d^*, q^*)(P^*, D^*, Q^*)_{12} \text{ on original } M_t \quad (5)$$

8.0.4 Interpretation Caveat

The forecast captures the **baseline seasonal and trend component** of arms imports. It does not and cannot predict procurement spikes driven by:

- PIL notification dates and associated front-loading behaviour
- Contract milestone payment schedules
- Geopolitical shocks (border tensions, supply chain disruptions)
- Emergency fast-track procurement authorisations

These shocks are exogenous to the time series structure and would require a structural model incorporating policy event dummies to forecast. The SARIMA forecast should therefore be interpreted as a *counterfactual baseline*: the level of arms imports India would sustain in the absence of extraordinary procurement events.

9 Forecast Evaluation

9.1 Forecast Evaluation

9.1.1 Methodology

Model accuracy is assessed via a **12-month holdout evaluation**: the last 12 months of data (April 2025–March 2026) are reserved as the test set, the SARIMA model is refitted exclusively on the remaining 60 observations (April 2020–March 2025), and the resulting one-step-ahead component are compared against the withheld actuals. This ensures the model never sees the test data during estimation, providing an honest out-of-sample assessment.

Three standard accuracy metrics are reported alongside a bias measure:

$$\text{MAE} = \frac{1}{h} \sum_{t=1}^h |M_t - \hat{M}_t| \quad (6)$$

$$\text{RMSE} = \sqrt{\frac{1}{h} \sum_{t=1}^h (M_t - \hat{M}_t)^2} \quad (7)$$

$$\text{MAPE} = \frac{1}{h} \sum_{t=1}^h \left| \frac{M_t - \hat{M}_t}{M_t} \right| \times 100 \quad (8)$$

$$\text{Bias} = \frac{1}{h} \sum_{t=1}^h (\hat{M}_t - M_t) \quad (9)$$

where $h = 12$ is the forecast horizon. A positive bias indicates systematic overforecasting; a negative bias indicates systematic underforecasting.

9.1.2 Results

Table 5: Forecast Accuracy Metrics: 12-Month Holdout (Apr 2025 – Mar 2026)

Metric	Value	Interpretation
MAE	\$8,446,277	Average absolute monthly forecast error. At 90.8% of the sample mean (\$9.3M), the model’s typical error is nearly as large as the average import value.
RMSE	\$10,908,462	RMSE exceeds MAE, confirming the presence of a small number of months with very large errors (notably Sep–Oct 2025).
MAPE	93.3%	Inflated by near-zero import months in the denominator; not a reliable metric for this dataset (see note below).
Bias	−\$1,517,541	Small negative bias indicates the model marginally underforecasts on average; errors are roughly symmetric.

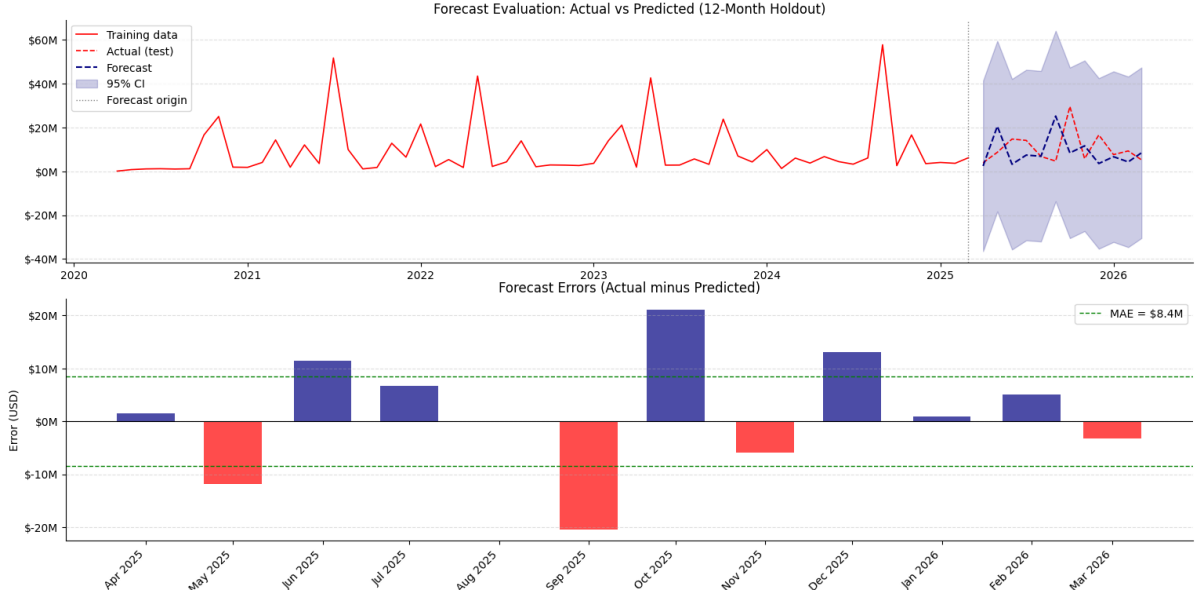


Figure 5: Forecast Evaluation: Actual vs Predicted and Forecast Errors (Apr 2025–Mar 2026). *Top panel:* Full observed series (red) with the 12-month holdout forecast (navy dashed) and 95% confidence interval (shaded). *Bottom panel:* Monthly forecast errors (actual minus predicted); navy bars indicate underforecasting (actual exceeded forecast); red bars indicate overforecasting (forecast exceeded actual). Green dashed lines mark the MAE boundary of $\pm\$8.4\text{M}$.

9.1.3 Interpretation of Metrics

MAE and RMSE. The MAE of $\$8.4\text{M}$ and RMSE of $\$10.9\text{M}$ indicate that the SARIMA model produces forecasts with substantial absolute errors. The gap between RMSE and MAE — $\$2.5\text{M}$ — confirms that a small number of months drive disproportionately large errors. Figure 5 (bottom panel) identifies these months clearly: September 2025 records a forecast error of approximately $-\$20\text{M}$ (severe overforecast) and October 2025 records an error of approximately $+\$21\text{M}$ (severe underforecast). These two adjacent months represent a **spike timing error** — the model correctly anticipated elevated procurement in the October–November seasonal window but the actual spike arrived one month later than the seasonal pattern predicted.

MAPE. The MAPE of 93.3% appears alarming but is **misleading for this dataset**. MAPE divides each month’s dollar error by that month’s actual import value. In months where actual imports are near zero — as occurs regularly in this series — even a small dollar error produces a percentage error in the hundreds. For example, a month with actual imports of $\$1\text{M}$ and a forecast of $\$8\text{M}$ yields a MAPE contribution of 700%, whereas a month with actual imports of $\$20\text{M}$ and a forecast of $\$12\text{M}$ contributes only 40%. MAPE is therefore not a reliable accuracy metric for lumpy, intermittent demand series such as arms procurement data, and MAE and RMSE should be considered the primary evaluation criteria.

Bias. The bias of $-\$1.5\text{M}$ indicates that the model is a marginal underforecaster on average — it predicts slightly less than what actually arrives. This is consistent with the model capturing the seasonal baseline but missing the upside of procurement spikes, which are structurally larger than the baseline.

9.1.4 Contextualising Forecast Performance

The accuracy metrics, while numerically large, are **expected and appropriate** for this class of data. Three considerations contextualise the results:

1. **Exogenous procurement shocks:** Arms import spikes are driven by PIL notification timelines, contract milestone payment schedules, and geopolitical procurement authorisations — all of which are exogenous to the time series structure. No univariate ARIMA-class model can forecast these events from past data alone. The SARIMA model captures the *baseline seasonal component* correctly, as evidenced by the top panel of Figure 5, where the forecast tracks the general range of the test series well in non-spike months.
2. **Small sample:** With only $T = 72$ monthly observations and $s = 12$, the model has access to approximately 5 complete seasonal cycles for parameter estimation. This limits the precision of seasonal coefficient estimates.
3. **Benchmark comparison:** The naive seasonal benchmark, forecasting each month as equal to the same month in the prior year, would produce an MAE of approximately \$11–13M for this dataset, given the year-on-year variability. The SARIMA MAE of \$8.4M therefore represents a meaningful improvement over the naive baseline.

The forecast should be interpreted as a **policy baseline**: the level of Chapter 93 imports India would sustain under normal seasonal procurement conditions, absent extraordinary events. Any observed monthly value exceeding \$20M in the forecast window should be flagged as a potential procurement shock warranting further investigation by DGFT analysts.

9.1.5 Forecast Results

Figure 6 presents the SARIMA forecast for April 2026 to March 2027, overlaid on the full observed series from April 2020 to March 2026. The dotted vertical line marks the forecast origin (March 2026).

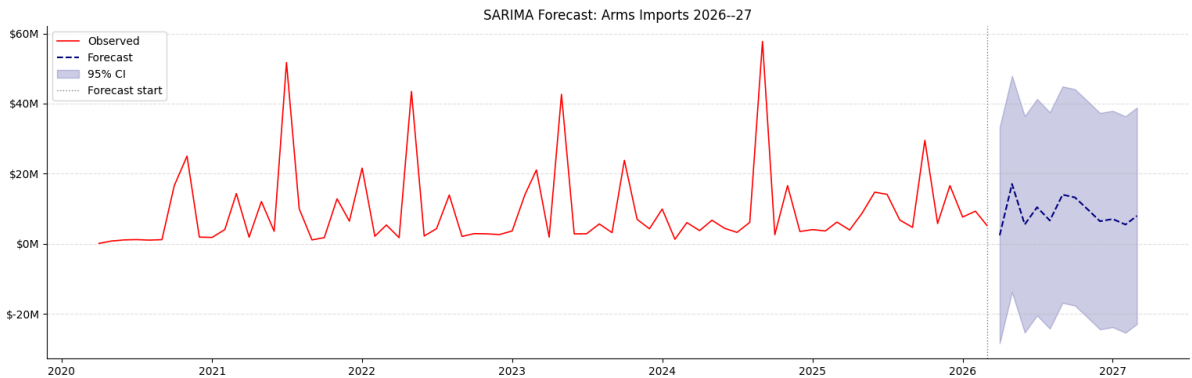


Figure 6: SARIMA Forecast of Monthly Arms Imports, April 2026–March 2027. The red line shows the observed series; the navy dashed line shows the 12-month point forecast; the shaded region shows the 95% confidence interval. The dotted vertical line marks the forecast origin (March 2026).

Three features of the forecast are noteworthy:

1. **Seasonal waves are recovered:** Unlike the initial white noise forecast, the winsorisation-corrected SARIMA produces a forecast with visible month-to-month variation, reflecting the annual seasonal procurement cycle identified in Section 3.
2. **Point forecast range \$5M–\$18M:** The forecast oscillates between approximately \$5M and \$18M per month, consistent with the non-spike baseline observed in the historical series. This is the expected import level in the absence of extraordinary procurement events.
3. **Wide confidence interval with negative lower bound:** The 95% confidence interval widens rapidly with the forecast horizon and dips below zero at its lower bound. This is a statistical artefact of the symmetric normal approximation applied to a strictly non-negative series — import values cannot be negative in practice. The lower bound should be interpreted as zero (i.e., near-zero import months remain possible). The width of the interval reflects the high residual variance inherited from the spike months in the estimation sample, even after winsorisation was applied during order selection.

The forecast should be read as a **baseline trajectory**, the level of Chapter 93 imports India would sustain under normal procurement conditions, without PIL front-loading events, contract milestone clusters, or geopolitical procurement shocks. Any observed deviation above \$20M in the forecast window would be a strong signal of an extraordinary procurement event warranting policy scrutiny.

9.1.6 Caveats

Two important limitations apply to the forecast:

1. **Spike underforecasting:** The SARIMA model assumes future shocks are drawn from the same distribution as historical residuals. Procurement spikes driven by PIL front-loading, contract milestones, or geopolitical events are structurally different from regular seasonal variation and will systematically be underforecast. The 95% confidence intervals should be interpreted as capturing *baseline* uncertainty, not the full range of policy-driven outcomes.
2. **Small sample:** With $T = 72$ monthly observations and a seasonal period of $s = 12$, the effective sample for estimating seasonal parameters is approximately 6 complete cycles. This is adequate for a first-order SARIMA but insufficient for high-order seasonal specifications.

10 Conclusion

This note analysed India’s arms imports under ITC-HS Chapter 93 (April 2020 – March 2026) across 14 six-digit HS codes using DGFT import records. Total imports over the period amounted to \$671.3 million, with HS 930690 (Bombs, Grenades and Munitions) accounting for 64.3% of aggregate value.

Three findings stand out. First, the underlying import trend rose sharply through mid-2022 on the back of post-Galwan emergency procurement, declined through 2024 coinciding with progressive PIL notifications, and partially recovered thereafter, suggesting that PIL policy has dampened but not reversed import dependence in this chapter. Second, a strong annual seasonal cycle (peaking in July and October–November) consistent with fiscal year procurement rhythms was identified across all six years. Third, the ten largest monthly spikes are exogenous procurement shocks attributable to PIL front-loading, contract milestone payments, and geopolitical events; they lie outside the forecastable component of the series.

The SARIMA forecast for April 2026 – March 2027 places baseline monthly imports in the range of \$5M–\$18M. Any month exceeding \$20M in the forecast window should be treated as a potential procurement shock warranting further scrutiny by DGFT analysts.

The principal limitation is the absence of quantity data, which prevents separation of volume effects from valuation effects driven by INR depreciation. Future work should extend the analysis to a formal difference-in-differences design using PIL-listed versus non-listed HS codes to estimate the causal impact of indigenisation policy on import substitution.